

MLP

Lab (9)

Case Study 1: Iris Species Classification

- **Objective: Multiclass classification using a classic dataset.**
- **Tasks:**
 - Train an MLPClassifier with 1 hidden layer (e.g., 10 neurons).
 - Evaluate accuracy, precision, and recall.
 - Visualize the decision boundary for two features (sepal length/width) using a mesh grid.

Case Study 2: Diabetes Progression Prediction

- **Objective: Regression with medical data.**
- **Tasks:**
 - Load the Diabetes dataset (`sklearn.datasets.load_diabetes`).
 - Split data (80% train, 20% test) and standardize features.
 - Train an MLPRegressor with two hidden layers (e.g., 20 and 10 neurons).
 - Evaluate performance using MSE and R^2 score.
 - Compare results with linear regression (benchmark).

Case Study 3: Wine Quality Classification

- **Objective: Imbalanced multiclass classification.**
- **Tasks:**
 - Load the Wine dataset (`sklearn.datasets.load_wine`).
 - Handle class imbalance (e.g., stratify splits, class weighting).
 - Train an MLPClassifier with ReLU activation and early stopping.
 - Generate a confusion matrix and classification report.
 - Perform feature importance analysis using permutation importance.

Case Study 4: Synthetic Non-Linear Boundary Prediction

- **Objective: Learn complex decision boundaries.**
- **Tasks:**
 - Generate a synthetic dataset (e.g., `make_moons` or `make_circles` with noise).
 - Split data and visualize original distribution.
 - Train an `MLPClassifier` with varying hidden layers (e.g., 5 vs. 50 neurons).
 - Plot decision boundaries for both architectures.
 - Discuss overfitting/underfitting by comparing test accuracy.

```
# 1. Generate data
X, y = make_moons(n_samples=1000, noise=0.2, random_state=42)
```

Case Study 5: Customer Churn Prediction

- **Objective: Binary classification with real-world tabular data.**
- **Tasks:**
 - Load the Telco Customer Churn dataset (from CSV).
 - Preprocess data: encode categorical variables, handle missing values, normalize.
 - Train an `MLPClassifier` (e.g., 32-neuron hidden layer, Adam solver).
 - Optimize hyperparameters (learning rate, alpha) via grid search.
 - Compute ROC-AUC score and plot the ROC curve.

```
# 1. Load data
url = "https://raw.githubusercontent.com/IBM/telco-customer-churn-on-icp4d/master/data/Telco-Customer-Churn.csv"
```